STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE – 10

APPLICATIONS OF PROBABILITY THEORY IN MACHINE LEARNING

NAIVE BAYE'S ALGORITHM & DECISION TREES

PREPARED BY
HAZBER SAMSON
FAST-NUCES ISLAMABAD

NAIVE BAYE'S ALGORITHM

Naive Bayes is a probabilistic classification algorithm based on Bayes' Theorem, which calculates the probability of a class given the input features.

It assumes that all predictors are conditionally independent of one another given the class label, which simplifies computation significantly.

This assumption enables the algorithm to work efficiently, even with high-dimensional data, by estimating probabilities for each feature individually.

Despite its simplicity, Naive Bayes is highly effective in many applications, such as spam filtering, text classification, and sentiment analysis, where it delivers robust and reliable results.

We shall discuss two versions of Baye's Theorem.

1- Baye's Theorem for Single Feature
2- Baye's Theorem for Multiple Features

BAYES THEOREM FOR SINGLE FEATURE

Suppose we have a class labels "yes" or "No" and a single feature "X".

Now Baye's Theorem for single feature X and having a class label "yes" (Y) is given by

$$P(Y|X) = \frac{P(X|Y).\ P(Y)}{P(X)}$$

And Baye's Theorem for single feature X and having a class label "No" (N) is given by

$$P(N|X) = \frac{P(X|N).\ P(N)}{P(X)}$$

BAYES THEOREM FOR MULTIPLE FEATURES

Suppose we have a class labels "yes" or "No" and multiple features $X_1, X_2, \ldots, X_n$.

Now Baye's Theorem for multiple features $X_1, X_2, \ldots, X_n$ and having a class label "yes" (Y) is given by

$$P(Y|X_1, X_2, \ldots, X_n) = \frac{P(X_1|Y).\ P(X_2|Y).\ldots P(X_n|Y).P(Y)}{P(X_1).P(X_2).\ldots P(X_n)}$$

Now Baye's Theorem for multiple features $X_1, X_2, \ldots, X_n$ and having a class label "yes" (Y) is given by

$$P(N|X_1, X_2, \ldots, X_n) = \frac{P(X_1|N).\ P(X_2|N).\ldots P(X_n|N).P(N)}{P(X_1).P(X_2).\ldots P(X_n)}$$

HAZBER SAMSON

EXAMPLES OF NAÏVE BAYE'S ALGORITHM

EXAMPLE-1 Consider the data of 10 persons indicating COVID , flu and fever. Use Naïve Bayes Algorithm to classify that a person taken at random will have fever or not. Use COVID and Flu as features.

Person	COVID (Yes/No)	Flu (Yes/No)	Fever (Yes/No)
1	Yes	No	Yes
2	No	Yes	Yes
3	Yes	Yes	Yes
4	No	No	No
5	Yes	No	Yes
6	No	No	Yes
7	Yes	No	Yes
8	Yes	No	No
9	No	Yes	Yes
10	No	Yes	No

SOLUTION We shall first calculate Prior Probabilities then Conditional Probabilities

STEP-1 PRIOR PROBABILITIES

$P(fever = yes) = 7/10 \quad and \quad P(fever = no) = 3/10$

STEP-2 CONDITIONAL PROBABILITIES

$P(flu \mid fever = yes) = 3/7 \quad and \quad P(\text{cov}id \mid fever = yes) = 4/7$

$P(flu \mid fever = no) = 2/3 \quad and \quad P(\text{cov}id \mid fever = no) = 2/3$

STEP-3 POSTERIOR PROBABILITIES

$$P(fever = yes \mid flu, \text{cov}id) = P(flu \mid fever = yes) * P(\text{cov}id \mid fever = yes) * P(fever = yes)$$

$$= \frac{3}{7} \times \frac{4}{7} \times \frac{7}{10} = 0.1714$$

$$P(fever = no \mid flu, \text{cov}id) = P(flu \mid fever = no) * P(\text{cov}id \mid fever = no) * P(fever = no)$$

$$= \frac{2}{3} \times \frac{2}{3} \times \frac{3}{10} = 0.13$$

CONCLUSION

Since $P(fever = yes \mid flu, \text{cov}id) > P(fever = no \mid flu, \text{cov}id)$

$\Rightarrow$ *it is predicted that a person selected at random having flu and* cov*id might have fever.*

HAZBER SAMSON

EXAMPLE-2 To Predict **Play Tennis (Yes/No)** based on the features **Outlook (Sunny/Overcast/Rain)** and **Humidity (High/Low)** using Naive Bayes, the probabilities of **Play Tennis = Yes** and **Play Tennis = No** are calculated using Bayes' Theorem, and the class with the higher probability is selected as the prediction.

SOLUTION

Day	Outlook	Humidity	Play Tennis
1	Sunny	High	No
2	Sunny	High	No
3	Overcast	High	Yes
4	Rain	High	Yes
5	Rain	Low	Yes
6	Rain	Low	No
7	Overcast	Low	Yes
8	Sunny	High	No
9	Sunny	Low	Yes
10	Rain	High	No

STEP-1 PRIOR PROBABILITIES

The **prior probability** is the probability of the target variable (`Play Tennis`) being `Yes` or `No` without considering any features.

P(Play Tennis = Yes)

Count how many times `Play Tennis = Yes`:

- `Yes` appears in rows 3, 4, 5, 7, and 9, so there are **5 instances** of `Play Tennis = Yes`.

$$P(\text{Play Tennis} = \text{Yes}) = \frac{5}{10} = 0.5$$

P(Play Tennis = No)

Count how many times `Play Tennis = No`:

- `No` appears in rows 1, 2, 6, 8, and 10, so there are **5 instances** of `Play Tennis = No`.

$$P(\text{Play Tennis} = \text{No}) = \frac{5}{10} = 0.5$$

STEP-2 CONDITIONAL PROBABILITIES

Next, we calculate the **conditional probabilities** for each feature given the target class (`Play Tennis = Yes` or `Play Tennis = No`).

For **Play Tennis = Yes** (rows 3, 4, 5, 7, 9):

1. P(Outlook = Sunny | Play Tennis = Yes):

Out of the 5 rows where `Play Tennis = Yes` (rows 3, 4, 5, 7, 9), the `Outlook = Sunny` appears **once** (in row 9).

$$P(\text{Outlook} = \text{Sunny} \mid \text{Play Tennis} = \text{Yes}) = \frac{1}{5} = 0.2$$

2. P(Humidity = High | Play Tennis = Yes):

Out of the 5 rows where `Play Tennis = Yes`, `Humidity = High` appears **twice** (rows 3 and 4).

$$P(\text{Humidity} = \text{High} \mid \text{Play Tennis} = \text{Yes}) = \frac{2}{5} = 0.4$$

For **Play Tennis = No** (rows 1, 2, 6, 8, 10):

1. P(Outlook = Sunny | Play Tennis = No):

Out of the 5 rows where `Play Tennis = No` (rows 1, 2, 6, 8, 10), the `Outlook = Sunny` appears **three times** (rows 1, 2, and 8).

$$P(\text{Outlook} = \text{Sunny} \mid \text{Play Tennis} = \text{No}) = \frac{3}{5} = 0.6$$

2. P(Humidity = High | Play Tennis = No):

Out of the 5 rows where `Play Tennis = No`, `Humidity = High` appears **four times** (rows 1, 2, 8, and 10).

$$P(\text{Humidity} = \text{High} \mid \text{Play Tennis} = \text{No}) = \frac{4}{5} = 0.8$$

STEP-3 POSTERIOR PROBABILITIES

Let's predict whether a person will play tennis given the following features:

- Outlook = Sunny
- Humidity = High

We'll calculate the posterior probabilities for both classes (`Play Tennis = Yes` and `Play Tennis = No`) using Bayes' Theorem.

Step 4.1: **Calculate Posterior Probability for** `Play Tennis = Yes`

Using Bayes' Theorem, the posterior probability is proportional to:

$$P(\text{Play Tennis} = \text{Yes} \mid \text{Features}) \propto P(\text{Outlook} = \text{Sunny} \mid \text{Play Tennis} = \text{Yes}) \cdot P(\text{Humidity} = \text{High} \mid \text{Play Tennis} = \text{Yes}) \cdot P(\text{Play Tennis} = \text{Yes})$$

We already know the values:

- P(Outlook = Sunny | Play Tennis = Yes) = 0.2
- P(Humidity = High | Play Tennis = Yes) = 0.4
- P(Play Tennis = Yes) = 0.5

So, the posterior probability for `Play Tennis = Yes` is:

$$P(\text{Yes} \mid \text{Features}) \propto 0.2 \cdot 0.4 \cdot 0.5 = 0.04$$

Step 4.2: **Calculate Posterior Probability for** `Play Tennis = No`

Now, calculate the posterior probability for `Play Tennis = No`:

$$P(\text{Play Tennis} = \text{No} \mid \text{Features}) \propto P(\text{Outlook} = \text{Sunny} \mid \text{Play Tennis} = \text{No}) \cdot P(\text{Humidity} = \text{High} \mid \text{Play Tennis} = \text{No}) \cdot P(\text{Play Tennis} = \text{No})$$

We already know the values:

- P(Outlook = Sunny | Play Tennis = No) = 0.6
- P(Humidity = High | Play Tennis = No) = 0.8
- P(Play Tennis = No) = 0.5

So, the posterior probability for `Play Tennis = No` is:

$$P(\text{No} \mid \text{Features}) \propto 0.6 \cdot 0.8 \cdot 0.5 = 0.24$$

STEP-4 MAKE A PREDICTION

Now, compare the posterior probabilities:

- **P(Play Tennis = Yes | Features) = 0.04**
- **P(Play Tennis = No | Features) = 0.24**

Since **P(Play Tennis = No | Features) = 0.24** is greater than **P(Play Tennis = Yes | Features) = 0.04**, the **Naive Bayes** classifier will predict that the person **will not play tennis**.

Final Answer:

The prediction is: Play Tennis = No.

HAZBER